

APPLICATIONS

**STATISTICAL EVALUATION OF SEQUENTIAL
DETONATION SYSTEMS**

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Summary: Detonators are extensively used in mining operations. Because of the great depths at which such operations are conducted, especially in Gold mines, proper performance of detonators is of critical importance. A common operating procedure is to insert a detonator into each of a series of shotholes drilled into a stoepface. The detonators are connected in series and are supposed to explode in sequence. Detonations which occur out of sequence can cause all manner of problems, among them difficulties in clearing the stoepface and premature termination of the blast, both of which involve potentially serious hazards to the safety of miners. Thus, an important statistical parameter of the system is the probability that shots will occur out of sequence. In this paper some characteristics of sequential detonation systems are studied from a statistical perspective. In particular, it is found that the standard deviation of detonator delay times, rather than their mean, is a critical parameter. This finding implies that the main thrust of the quality assurance effort should be directed towards the more difficult problem of controlling the standard deviation of manufactured delays rather than at the relatively easy problem of controlling their mean. Simple approximate formulas are provided for estimating the probability that shots will occur out of sequence.

1. Introduction

Sequential blasting systems are widely used in underground and surface mining operations. The basic ingredients of such a system are depicted in schematic form in figure 1. Successive main delay detonators are inserted into shotholes drilled in a stopeface. These are connected to a series of surface delay detonators via shock tubing or, in older systems, by ignitercord. The delays consist of lengths of pyrotechnic material which are supposed to burn at a more or less uniform rate. Consider, for instance, a sequence of 4 000 millisecond (*ms.*) main delays joined by 200 *ms.* surface delays. If the sequence is initiated at point A in figure 1 one expects successive detonations to occur after 4 000, 4 200, 4 400, . . . *ms.* While the first main delay in the sequence burns, successive main delays are activated every 200 *ms.* via the surface delays. After 4 000 *ms.*, when the first detonation occurs, the next $4\,000/200 = 20$ main delays are already active. The first of these has burned for 3 800 *ms.*, the next for 3 600 *ms.*, and so on up to the twentieth which has just been initiated. This series of initiated main delays is the so-called initiation front. A detonation can easily result in damage to surface delays in its proximity, for example by rocks falling from a stopeface overhang. If the initiation front has not developed beyond such a point it could terminate there, resulting in an incomplete blasting operation. Thus, a long initiation front is desirable. The connecting delays create a time lapse between successive detonations which is necessary for an economically efficient and a safe blasting operation. In the preceding deterministic setup it is a trivial matter to design a fully efficient blasting operation. If, say, one hundred detonators are to be used with a

required initiation front of 20 and a 200 *ms.* time lapse between detonations, one would simply use a series of one hundred 4 000 *ms.* main delays connected by ninety-nine 200 *ms.* surface delays. Such is indeed the case with certain electronic systems. These are, however, relatively expensive to operate with the result that by far the greater share of the detonator market is currently held by pyrotechnic-based systems.

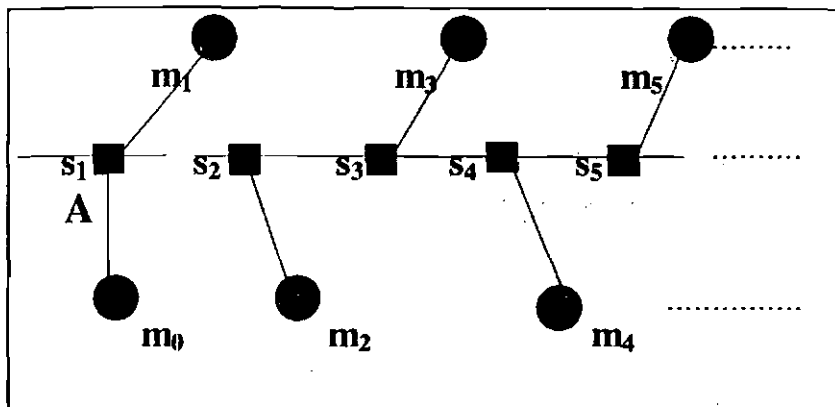


Figure 1. Schematic view of blast pattern often used in Gold mining. The solid circles represent shotholes containing main delays, the squares surface delays. The connecting lines represent the shock tube or ignitercord.

Thus, the situation is complicated by the fact that burning times of delays are subject to considerable inherent statistical variation. This variation leads to a

phenomenon, called reversal, in which detonations occur out of their natural sequence such as when, for example, the fourth in the sequence detonates before the third. The occurrence of such reversal events has potentially serious consequences in, especially, underground mining operations. Thus it is in the interest of both the manufacturer and the end user of such detonation systems to be in a position to estimate the probability of such events occurring and to be aware of the effect statistical variability has on the frequency of their occurrence. Our experience in the industry indicates that manufacturers and users of explosives, while no doubt aware of the importance of the reversal probability, are largely unaware of the relationship between various blast parameters and statistical measures of blast efficiency. There seems to be a widely held belief that adequate control of the mean delay is sufficient to guarantee a small reversal probability. We will see that this belief is mistaken and that control of the variability of delay times is just as, if not more, important than control of the mean.

Trollip *et al.* (1967) discussed certain aspects of the reversal problem. They consider only "pairwise" reversal probabilities, that is, the probability that number $i + 1$ in the sequence will detonate before number i . Our approach is more general in that a formula is given for calculating the probability that an array of any given length will not detonate in sequence. This probability is substantially larger than the pairwise reversal probability, the latter being a special case pertaining to arrays consisting of two detonators only. A search of the generally accessible mining and statistical literature yielded no further references to the specific topic being addressed in the present paper.

Here is an outline of the contents of our paper. In section 2 a statistical model for delay time data, assumed normally distributed, is formulated and the basic parameters of interest, namely the reversal probability and the initiation front, are considered. Expressions are derived for the reversal probability and for the distribution of the initiation front. These clearly indicate the nature of the dependence of the basic parameters on the mean and variance of delay times. An interesting point is that reversal probability does not depend directly on the mean of the main delay times, but rather on their standard deviation. Also, the length of the initiation front is essentially determined by the ratio of main to surface delay times, the variability of the latter being extremely small in most cases of practical interest. An implication of these findings is that quality control procedures should focus as much, if not more, on control of the standard deviation of main delay times as on their mean.

Section 3 considers estimation of the basic parameters from data gathered as part of routine quality control operations. A point and interval estimate of reversal probability is proposed and their large and small sample properties are investigated. Monte Carlo experiments indicate that the coverage probabilities of the confidence intervals are very close to their nominal values even in small samples. Section 4 illustrates the application of the methodology to a set of quality control data.

Section 5 considers the same problem for data obtained in field tests conducted under operational conditions. Here the available data typically consist only of a sequence of times at which the respective detonations occurred, individual delay times being unobservable. An interesting point concerns the rather peculiar auditory pattern of explosions that seems to characterize

most blasting operations. This effect tends to be perceived in a negative light by, especially, prospective purchasers of the product to whom its capabilities are being demonstrated. It is shown that the effect is intrinsic to any non-deterministic sequential blasting system.

In section 6 we illustrate the effect of non-normally distributed delays on the reversal probability. Especially, if the tails of the delay distribution are heavier than those of a normal distribution then failure to take account of this in the calculations can lead to gross underestimation of the reversal probability. We also suggest a simple, computer intensive method of calculating the reversal probability when the normality assumption is violated.

Finally, in section 7 we illustrate the application of the method developed in this paper to a detonator configuration other than that shown in figure 1, namely one typically used in Platinum mines. It will be seen that once the concept of "reversal" has been properly defined in respect of any given configuration, a simple and very general formula can be applied to calculate the reversal probability. Proofs and technical details relating to certain calculations are given in an appendix.

2. Basic parameters

2.1 The reversal probability

The main delays $m_0, \dots, m_n$ and surface delays $s_1, \dots, s_n$ depicted in figure 1 will be assumed statistically independent $N(\mu_m, \sigma_m^2)$ and $N(\mu_s, \sigma_s^2)$ respectively, where the notation $N(a, b^2)$ denotes a normal (Gaussian) distribution with mean a and variance b^2 . It is important to keep

in mind that the standard deviation of a delay increases with its mean. Figure 2 shows a scatterplot of $\log(\text{standard deviation})$ against $\log(\text{mean})$ of a number of manufactured batches of delays. The nature of the dependence could be of the form

$$\text{standard deviation} = c \times \text{mean}^\beta \quad (1)$$

(with substantial added noise) where c and β are positive constants which depend on the physical properties of the pyrotechnic material used. A typical set of parameters would be, for example, $\mu_m = 4\,000$, $\sigma_m = 50$ and $\mu_s = 200$, $\sigma_s = 2$. In particular, the standard deviation is typically two orders of magnitude less than the mean while the surface delay parameters μ_s and σ_s are typically an order of magnitude smaller than their respective main delay counterparts μ_m and σ_m . In order to simplify certain calculations we will from time to time make the approximation $\sigma_s/\sigma_m \approx 0$. If it were possible to manufacture a 4 000 ms. pyrotechnic delay which has the same small standard deviation as a 200 ms. surface delay, the reversal phenomenon would, of course, essentially disappear.

Let T_j denote the time instant at which detonator number j ($= 0, \dots, n$) explodes. The detonators explode in sequence if and only if each of the events $\{T_{k-1} \leq T_k\}$, $k = 1, \dots, n$, occurs. The complement of this event is

$$R = \cup_{k=1}^n \{T_{k-1} > T_k\} \quad (2)$$

and the reversal phenomenon will be observed only when R occurs. The probability, $\Pr\{R\}$, that R occurs, is one of the basic parameters and will be referred to as the *reversal probability*.

Let A_k denote the event $\{T_{k-1} > T_k\}$ and set $\theta = \Pr\{A_k\}$. This θ , which will shortly be shown to be independent of k , is referred to as the *pairwise reversal probability*. The time T_j can be expressed as

$$T_j = \begin{cases} m_0 & ; j = 0 \\ s_1 + \dots + s_j + m_j & ; 1 \leq j \leq n. \end{cases} \quad (3)$$

It follows that the event A_k can be expressed as $A_k = \{s_k + m_k - m_{k-1} < 0\}$. Since $s_k + m_k - m_{k-1}$ is $N(\mu_s, \sigma_s^2 + 2\sigma_m^2)$, independently of k , we see that

$$\theta = \Phi(x) \quad \text{with} \quad x = -\mu_s/(\sigma_s^2 + 2\sigma_m^2)^{1/2} \quad (4)$$

and where Φ denotes the distribution function of the $N(0,1)$ distribution, that is

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(-t^2/2) dt. \quad (5)$$

Notice that θ does not depend on the mean main delay time μ_m . Tables 1a and 1b, which give some pairwise reversal probabilities when $\mu_s = 0.2$ s. and 0.5 s. respectively, are intended to give the reader some feeling for the orders of magnitude of the probabilities involved, and of the extent of their dependence on σ_s and σ_m . Clearly, proportional changes in the main delay standard deviation have a much larger effect than corresponding proportional changes in surface delay standard deviations.

Table 1a. Pairwise reversal probabilities θ for $\mu_s = 200$ and various combinations of σ_s and σ_m .

		σ_m			
		30	40	50	60
σ_s	4	1.34×10^{-6}	2.10×10^{-4}	2.37×10^{-3}	9.28×10^{-3}
	6	1.52×10^{-6}	2.19×10^{-4}	2.41×10^{-3}	9.36×10^{-3}
	8	1.81×10^{-6}	2.32×10^{-4}	2.47×10^{-3}	9.47×10^{-3}
	10	2.23×10^{-6}	2.49×10^{-4}	2.55×10^{-3}	9.62×10^{-3}

Table 1b. Pairwise reversal probabilities θ for $\mu_s = 500$ and various combinations of σ_s and σ_m .

		σ_m			
		90	120	150	180
σ_s	12	4.59×10^{-5}	1.65×10^{-3}	9.30×10^{-3}	2.49×10^{-2}
	18	5.02×10^{-5}	1.70×10^{-3}	9.42×10^{-3}	2.50×10^{-2}
	24	5.66×10^{-5}	1.77×10^{-3}	9.59×10^{-3}	2.53×10^{-2}
	30	6.58×10^{-5}	1.86×10^{-3}	9.80×10^{-3}	2.55×10^{-2}

Table 2a. Reversal probabilities Θ for $\mu_s = 200$ and sequence length $n = 100$.

		σ_m			
		30	40	50	60
σ_s	4	1.3×10^{-4}	2.1×10^{-2}	2.1×10^{-1}	6.1×10^{-1}
	6	1.5×10^{-4}	2.2×10^{-2}	2.1×10^{-1}	6.1×10^{-1}
	8	1.8×10^{-4}	2.3×10^{-2}	2.2×10^{-1}	6.1×10^{-1}
	10	2.2×10^{-4}	2.5×10^{-2}	2.3×10^{-1}	6.2×10^{-1}

Table 2b. Reversal probabilities Θ for $\mu_s = 500$ and sequence length $n = 100$.

		σ_m			
		90	120	150	180
σ_s	12	4.6×10^{-3}	1.5×10^{-1}	6.1×10^{-1}	9.2×10^{-1}
	18	5.0×10^{-3}	1.6×10^{-1}	6.1×10^{-1}	9.2×10^{-1}
	24	5.6×10^{-3}	1.6×10^{-1}	6.2×10^{-1}	9.2×10^{-1}
	30	6.6×10^{-3}	1.7×10^{-1}	6.3×10^{-1}	9.2×10^{-1}

Pairwise reversal probabilities are of little interest per se since they refer only to the probability that a reversal occurs in any given pair of adjacent detonations. From a practical point of view a more important quantity is the (substantially larger) probability that a reversal occurs *somewhere along the line* in a sequence of detonations, that is the probability of reversal $\Pr\{R\}$ (see (2)) in a sequence

of given length $n + 1$. If the A_k were statistically independent events, we would have

$$\Pr\{R\} = 1 - (1 - \theta)^n.$$

Now A_k and A_l are statistically independent whenever $|k - l| > 1$ but A_k and A_{k+1} are dependent since the main delay m_k figures in both. Nonetheless, it turns out that this dependence is so slight as to be of no practical consequence. The values $1 - (1 - \theta)^n$ corresponding to the θ values in tables 1a and 1b are displayed in tables 2a and 2b below for the case $n = 100$. (The case $n = 100$ is used purely for illustrative purposes. Substantially longer sequence lengths can occur in practice.) Direct calculation of the *exact* values of $\Pr\{R\}$, outlined in section 1 of the appendix, showed that the approximations to $\Pr\{R\}$ given in tables 2a and 2b are accurate to at least the number of significant digits quoted. Consequently, in what follows we focus attention on the quantity

$$\Theta = 1 - (1 - \Phi(x))^n, \quad (6)$$

treating it as if it were, in fact, the reversal probability.

2.2 The initiation front

Now we consider the length of the initiation front, I , which equals the number of main delays in the sequence ahead of the first (in a conceptually infinite sequence) that have already been initiated at the instant when the first detonation occurs. Mathematically, for $j = 1, 2, \dots$

$$I = j \text{ when } t_j \leq m_0 \text{ but } t_{j+1} > m_0$$

where

$$t_j = s_1 + \dots + s_j$$

denotes the time instant at which the j th detonator in the sequence is initiated ($t_0 = 0$). The reader should note that there is in fact a sequence of lengths $I_0, I_1, \dots$ of initiation fronts, one associated with each detonator in the sequence. It is desirable that the average initiation front be large and that its length be stable. It is clear, however, that the initiation fronts are highly correlated and statistically identical. For this reason attention can be restricted to a consideration of the first one in the sequence, which is denoted by I . Then it is required that the mean initiation front μ_I be large and, for stability, that the standard deviation σ_I be small.

Simple arithmetic suggests that the mean initiation front μ_I should be about $\nu = \mu_m/\mu_s$. Mathematically, ν can be made as large as desired simply by choosing μ_m sufficiently large or μ_s sufficiently small. The latter is not a viable option since, as mentioned in the introduction, a certain minimum time lapse between successive detonations is an operational requirement. A constraint which limits the freedom of choice of ν is the empirically established fact, mentioned earlier, that the larger the mean delay the larger also the standard deviation; see (1) and figure 2. Consider, for instance, a situation where $\beta = 1$, $c = 0.01$ in (1). Making the approximation $\sigma_s/\sigma_m \approx 0$ in (4) one sees that $x \approx 100/\nu$ and $\theta \approx \Phi(-100/\nu)$. For $\nu = 150$, the optimal initiation front if a sequence of 150 detonators are to be fired in sequence, one then has $\theta = \Phi(-0.667) = 0.252$. This is an unacceptably high pairwise

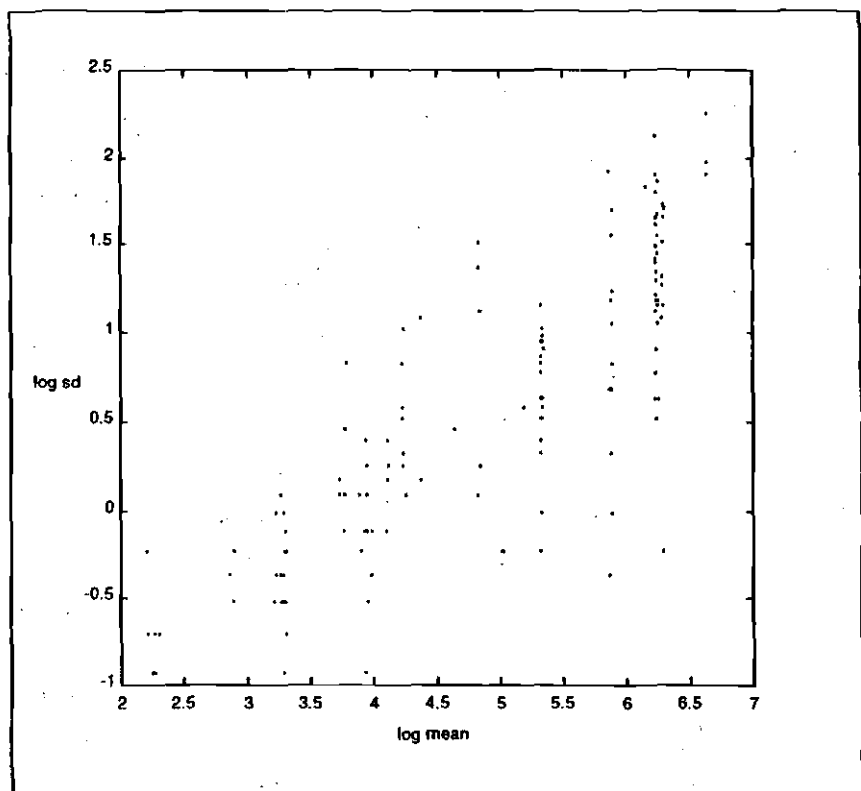


Figure 2. Relationship between mean and standard deviation of batches of delays.

reversal probability since it implies an overall reversal probability. $\Theta = 1$ for sequences of length $n > 30$. For later use we also note that in this instance

$$\sigma_m^2 + \nu\sigma_s^2 = \sigma_m^2(1 + \sigma_s/\sigma_m) \approx \sigma_m^2. \quad (7)$$

Analogous results hold for other choices of β but the calculations are different.

From the definition of the variable I ,

$$\begin{aligned} \Pr\{I \geq k\} &= \Pr\{s_1 + \dots + s_k \leq m_0\} \\ &= \Phi((\mu_m - k\mu_s)/(\sigma_m^2 + k\sigma_s^2)^{1/2}) \end{aligned} \quad (8)$$

for $k \geq 1$. The mean and standard deviation of I can therefore be found numerically to any desired degree of accuracy. General conclusions about the distribution of I can, however, be drawn from (8) without resorting to extensive numerical calculations. Suppose that $\nu = \mu_m/\mu_s$ is an integer. Then one can see from (8) that

$$\Pr\{I = r\} = 0.5 - \Phi(-\mu_s/(\sigma_m^2 + (\nu + 1)\sigma_s^2)^{1/2}) \approx 0.5$$

while

$$\Pr\{I = r - 1\} = 0.5 - \Phi(-\mu_s/(\sigma_m^2 + (\nu - 1)\sigma_s^2)^{1/2}) \approx 0.5.$$

The approximations follow because, from (7),

$$\Phi(-\mu_s/(\sigma_m^2 + (\nu \pm 1)\sigma_s^2)^{1/2}) \approx \Phi(-\mu_s/\sigma_m) \approx \Phi(-x)$$

is negligible compared to 0.5. Thus the initiation front effectively has length ν or $\nu - 1$, each with equal probability (which means that the standard deviation of the initiation front is $\sigma_I \approx 1/2$). It follows that the length of the initiation front is essentially determined by the ratio μ_m/μ_s of mean main delay to mean surface delay and is affected little by variations in the standard deviations of the delays. This stability is in sharp contrast to the effects of main delay

mean and standard deviation on reversal probability. Similar conclusions hold even if μ_m/μ_s is not an integer and when the assumption that all distributions are normal is not satisfied; see section 2 of the appendix.

3. Estimation from quality control data

Practical application of the formulae given in the preceding section requires knowledge of the unknown parameters μ_m , μ_s , σ_m and σ_s . Estimates of these will usually be obtained from data gathered as part of routine quality control operations. Suppose samples $s^{(1)}, \dots, s^{(k)}$ and $m^{(1)}, \dots, m^{(l)}$ of delay times are available and that $\hat{\mu}_s$, the sample mean surface delay, and $\hat{\sigma}_s^2$ and $\hat{\sigma}_m^2$, the sample variances of the surface and main delays, have been calculated. An obvious estimate of x in (4) is

$$\hat{x} = -\hat{\mu}_s / (\hat{\sigma}_s^2 + 2\hat{\sigma}_m^2)^{1/2} \quad (9)$$

leading, from (6) to the estimate $\hat{\Theta} = 1 - (1 - \Phi(\hat{x}))^n$ of Θ . Application of the "delta method" (see section 3 of the appendix for details) shows that $\hat{x}$ is, for large l , approximately normally distributed with mean x and variance $x^2 \times (2l)^{-1}$. The distribution of $\log(-\hat{x})$ will then be approximately normal with mean $\log(-x)$ and variance $(2l)^{-1}$, independent of x , and in small samples the approximation to normality of $\log(-\hat{x})$ can be expected to be superior to that of $\hat{x}$. Thus, an approximate 100β % confidence interval for x is given by

$$[L, U] := \left[\hat{x} \exp(z_{1-\beta/2} \times (2l)^{-1/2}), \hat{x} \exp(-z_{1-\beta/2} \times (2l)^{-1/2}) \right] \quad (10)$$

where $\Phi(z_{1-\beta/2}) = 1 - \beta/2$ and Φ is given in (5). Since $1 - (1 - \theta)^n$ is a strictly increasing function of x it follows from (10) and (6) that an approximate $100\beta\%$ confidence interval for $\Pr\{R\}$ is given by

$$[1 - (1 - \Phi(L))^n, 1 - (1 - \Phi(U))^n]. \quad (11)$$

The two intervals (10) and (11) will, of course, have exactly the same coverage probabilities. Table 3 presents some results of a Monte Carlo study undertaken to assess the coverage properties of the interval (11) for small sample sizes. Eight parameter and sample size configurations, shown in the first four columns, were used. One thousand Monte Carlo trials were performed in each case. The last three columns in table 3 show the empirical coverage probabilities of the interval (11) for nominal values $\beta = 0.90, 0.95$ and 0.99 .

As can be seen, overall agreement between empirical and nominal values is excellent. The *precision* with which Θ can be estimated will generally be unacceptably low for small sample sizes such as $l = 50$. For instance, the average width of the thousand 90% confidence intervals corresponding to the first row of table 3 is 0.192 whereas the reversal probability being estimated is $\Theta = 0.021$. However, if one regards reversal probability as a process characteristic rather than a property of an individual batch of manufactured product then it is natural to combine quality control data from a substantial number of more or less homogeneous batches of product for the purpose of estimating reversal probability. Then the effective sample size l is substantial (often of the order of 500 or more) and estimating precision should then be at an acceptable level; see section 4 below for an illustration.

Table 3. Empirical coverage probability of the confidence interval for reversal probability Θ in sequences of length 101 for nominal values $\beta = 0.90, 0.95$ and 0.99 based on 1 000 Monte Carlo trials.

μ_s	σ_s	σ_m	$k = l$	Θ	$\beta = 0.90$	$\beta = 0.95$	$\beta = 0.99$
200	2	40	50	2.1×10^{-2}	0.87	0.93	0.98
200	2	60	50	6.1×10^{-1}	0.90	0.95	0.99
200	4	40	50	2.1×10^{-2}	0.89	0.94	0.99
200	4	60	50	6.1×10^{-1}	0.90	0.94	0.98
500	5	80	50	5.1×10^{-4}	0.89	0.94	0.99
500	5	120	50	1.5×10^{-1}	0.89	0.94	0.99
500	10	80	50	5.4×10^{-4}	0.91	0.96	0.99
500	10	120	50	1.5×10^{-1}	0.88	0.93	0.98

Since $\Theta = 1 - (1 - \Phi(x))^n$ is a highly nonlinear function of x , it is to be expected that its natural point estimate $\hat{\Theta} = 1 - (1 - \Phi(\hat{x}))^n$ could be severely biased when l is small but that the bias will decrease as l increases. One way of reducing bias is to apply a two-sample jackknife; see Cox and Hinkley (1974, page 264). However, an unbiased estimate of unacceptably low precision is of no practical use. With $l = 500$ the bias in the raw estimate $\hat{\Theta}$ should be negligible; see section 4 for an illustration.

4. An example

Table 4 gives the means and standard deviations of samples of fifty main delays taken from each of twenty consecutive batches of manufactured product. The manufacture was controlled to meet target values $\mu_m = 4\,500$ and $\sigma_m \leq 40$. The question to be answered was: "Given these data, what would the reversal probability be in sequences of length 150 if the output of the production process

were used in conjunction with surface delays of known mean and standard deviation $\mu_s = 200$ and $\sigma_s = 2$ respectively?"

Table 4. Means and standard deviations of twenty samples consisting of fifty main delays each.

batch no.	mean	std.dev.
1	4503.5	35.2
2	4513.7	36.5
3	4492.9	29.7
4	4491.5	40.0
5	4503.7	47.0
6	4510.3	36.8
7	4499.1	28.4
8	4482.8	31.2
9	4498.8	34.4
10	4501.8	32.0
11	4501.9	33.3
12	4502.4	39.1
13	4499.8	28.3
14	4490.7	34.6
15	4519.2	37.1
16	4508.3	35.6
17	4509.4	35.5
18	4501.3	39.8
19	4407.4	35.6
20	4505.1	31.8

Normal quantile plots of the fifty observations in each batch confirmed that the normality assumption is reasonable. Further analysis of the data showed no significant time trends in either batch means or standard deviations.

The mean and standard deviation of the pooled data set consisting of 1 000 observations are $\hat{\mu}_m = 4501.7 \text{ ms.}$ and $\hat{\sigma}_m = 36 \text{ ms.}$ respectively. Then $\hat{x} = -3.929$ yielding an estimate $\hat{\theta} = 4.3 \times 10^{-5}$ for the pairwise reversal probability. Table 5 gives the estimate of reversal probability and the 90 % confidence interval for sequence lengths 100, 150 and 200 as computed from the formulae in (6) and (11). Since the point estimate lies close to the centre of the confidence interval in every case, no appreciable bias should be expected. Indeed, the jackknifed estimates differ from the raw estimates by at most 0.0006. It is of some interest to note that while the reversal probability for a sequence length $n + 1 = 150$ with a main delay standard deviation $\sigma_m = 36 \text{ ms}$ is 0.0064, with $\sigma_m = 40 \text{ ms.}$ (the upper limit controlled for in this specific manufacturing process) it becomes 0.0304 - an almost fivefold increase. This points to the extreme sensitivity of the reversal probability to relatively small variations in the standard deviation of main delays.

Table 5. Point estimates and 90 % confidence limits for reversal probability Θ for three sequence lengths.

n	$\hat{\Theta}$	$\hat{\Theta}_J$	90 % confidence limits
100	0.0043	0.0039	[0.0023 , 0.0076]
150	0.0064	0.0059	[0.0034 , 0.0114]
200	0.0085	0.0085	[0.0046 , 0.0151]

5. Field-test data

5.1 Estimation

In most field tests of detonation systems only the times $T_1 - T_0, \dots, T_n - T_0$ are observable. Such data can, for instance, be obtained by replaying a video tape of the test. Unless the product in question is of poor quality it is unlikely that one or more reversals will be observed in a field test. Thus, while the event in question, namely reversal, has not occurred one would nevertheless want to estimate the probability of its occurrence from the available data. The time lapses $X_j = T_j - T_{j-1} = m_j + s_j - m_{j-1}$ have a common mean μ_s and covariances

$$\text{cov}(X_k, X_l) = \begin{cases} 2\sigma_m^2 + \sigma_s^2 & \text{for } k = l \\ -\sigma_m^2 & \text{for } |k - l| = 1 \\ 0 & \text{for } |k - l| \geq 2. \end{cases} \quad (12)$$

The moment estimators of μ_s , σ_m^2 and $\sigma^2 := 2\sigma_m^2 + \sigma_s^2$ are therefore

$$\hat{\mu}_s = n^{-1} \sum_{j=1}^n X_j = (T_n - T_0)/n,$$

$$\hat{\sigma}_m^2 = -n^{-1} \sum_{j=0}^{n-1} (X_j - \hat{\mu}_s)(X_{j+1} - \hat{\mu}_s) \text{ and}$$

$$\hat{\sigma}^2 = n^{-1} \sum_{j=1}^n (X_j - \hat{\mu}_s)^2$$

and we can estimate x and Θ by $\hat{x} = -\hat{\mu}_s/\hat{\sigma}$ and $\hat{\Theta} = 1 - (1 - \Phi(\hat{x}))^n$ respectively. Again, using the delta method, it can be shown that $\log(-\hat{x})$ is approximately normal with mean $\log(-x)$ and variance $(4n/3)^{-1}$. Then (11), with $2l$ replaced by $4n/3$, is an approximate $100\beta\%$ confidence interval for Θ . Also, any bias in $\hat{x}$ is removed effectively by calculating estimates

$\hat{x}_{(1)}$ and $\hat{x}_{(2)}$ from the two data segments $T_0, \dots, T_m$ and $T_{m+1}, \dots, T_n$ respectively and estimating x by $2\hat{x} - (\hat{x}_{(1)} + \hat{x}_{(2)})/2$. Here m denotes the integer part of $(n+1)/2$. Note that the dependence structure of the data precludes use of the usual "leave one out at a time" jackknife. Getting rid of the bias in $\hat{\Theta}$ remains a problem, though.

5.2 Irregular detonations

Field tests are sometimes conducted with a view to demonstrating a product's capabilities to a prospective client. While the client expects to hear detonations occurring at regular intervals, a distinct irregularity in the pattern of detonation is invariably heard. This irregularity may give rise to some scepticism on the client's part and to a large measure of irritation on the part of the engineers involved. There is a simple explanation for this phenomenon. For, it follows from (12) that the correlation between X_j and X_{j+k} is zero for $k \geq 2$ while that between X_j and X_{j+1} is

$$\rho = -\frac{\sigma_m^2}{\sigma_s^2 + 2\sigma_m^2}$$

which is always negative. In fact, since $\sigma_s < \sigma_m$ it follows that $-1/2 \leq \rho < -1/3$. Thus a shorter (longer) than average time lapse tends to be followed by one which is longer (shorter) than average. A rather distinct detonation pattern will therefore *always* manifest itself.

6. The effect of non-normality

Suppose the distribution of main delays is a mixture of two normal distributions, $N(\mu_m, \eta^2)$ and $N(\mu_m, \tau^2)$, in the proportions $\alpha : 1 - \alpha$ for some

$0 < \alpha < 1$. Such a distribution has heavier than normal tails and will be used here to illustrate the effect of tail thickness of the main delay distribution on reversal probability. (We continue to assume that the shorter surface delays are normally distributed.) The pairwise reversal probability for such data is

$$\begin{aligned}\lambda &= \Pr\{s_1 + m_1 - m_0 < 0\} \\ &= \alpha^2 \Phi\left(-\mu_s / (\sigma_s^2 + 2\eta^2)^{1/2}\right) + (1 - \alpha)^2 \Phi\left(-\mu_s / (\sigma_s^2 + 2\tau^2)^{1/2}\right) \\ &\quad + 2\alpha(1 - \alpha) \Phi\left(-\mu_s / (\sigma_s^2 + \eta^2 + \tau^2)^{1/2}\right).\end{aligned}$$

This λ is to be compared with the value $\theta = \Phi\left(-\mu_s / (\sigma_s^2 + 2\sigma_m^2)^{1/2}\right)$ that would be obtained if the delay distribution were normal with exactly the same mean μ_s and variance $\sigma_m^2 = \alpha\eta^2 + (1 - \alpha)\tau^2$ as the mixture distribution. For instance, with a surface delay mean and standard deviation $\mu_s = 200$, $\sigma_s = 4$ and with mixture distribution parameters $\mu_m = 4\,000$, $\eta = 35$, $\tau = 70.1$ and $\alpha = .9$ the variance of main delays is 40^2 and the pairwise reversal probability is $\lambda = 1.2 \times 10^{-3}$. For $N(4\,000, 40^2)$ distributed main delays, however, the pairwise reversal probability is $\theta = 2.1 \times 10^{-4}$. Then, for sequences of length $n = 100$ the reversal probability for the mixture data is $1 - (1 - \lambda)^{100} = 11.3 \times 10^{-2}$ while the corresponding figure for normal data is more than five times smaller, namely $1 - (1 - \theta)^{100} = 2.1 \times 10^{-2}$. Clearly, calculating the reversal probability as if the data were normal when, in reality, they are not can produce very misleading results. Thus, it is important to check continuously the conformance to normality of (especially the tails) of the main delay distribution if formulas (4) and (6) are to be used on a routine basis.

How can one calculate the reversal probability for a batch of manufactured product if the main delay distribution shows substantial deviations from normality such as heavier than normal tails or right-skewedness? The pairwise reversal probability is $\Pr\{s_1 + m_1 - m_0 < 0\}$ and conditioning on m_0 and m_1 gives $\theta = E[\Phi((- \mu_s + m_0 - m_1)/\sigma_s)]$. Hence, if main delay data $m^{(1)}, \dots, m^{(l)}$ are available one could estimate θ by

$$\hat{\theta} = l^{-2} \sum_{i=1}^l \sum_{j=1}^l \Phi((- \mu_s + \hat{m}_i - \hat{m}_j)/\sigma_s)$$

which requires l^2 lookups of the $N(0, 1)$ distribution function. This is a feasible proposition on a PC running FORTRAN or MATLAB or some such software, even for values of l as large as 100.

7. Another detonator configuration

Figure 3 is a schematic representation of a diamond-shaped configuration often used in Platinum mines. Number the detonators $0a, 0b, 1, 2a, 2b, 3 \dots$ and denote the times at which they explode by $T_{0a}, T_{0b}, T_1, T_{2a}, T_{2b}, T_3, \dots$. Let C_k denote the event $\max(T_{(2k-2)a}, T_{(2k-2)b}) < T_{2k-1} < \min(T_{(2k)a}, T_{(2k)b})$, $k = 1, \dots, n/2$ (n is assumed to be even). Then, for example, C_1 denotes that the *last* of $0a$ and $0b$ to explode does so before 1 which in turn explodes before the *first* of $2a$ and $2b$. Explosions are deemed to have occurred in sequence only when $C_1, \dots, C_{n/2}$ all occur. Again, the dependence between any two adjacent C 's is very slight so that the reversal probability is again approximately equal to $(1 - \theta)^n$ where $\theta = 1 - \Pr\{C_1\}$. It remains to find an expression for θ in terms of the delay

distribution parameters. A straightforward calculation, outlined in section 4 of the appendix, yields

$$\theta = 1 - E \left[\frac{\Phi^2(\mu_s/\sigma_m + \min(Z_{1a}, Z_{1b})\sigma_s/\sigma_m + Z_1)}{\times \Phi^2(\mu_s/\sigma_m + Z_2\sigma_s/\sigma_m - Z_1)} \right]$$

where Z_{1a} , Z_{1b} , Z_1 and Z_2 denote independent $N(0, 1)$ random variables.

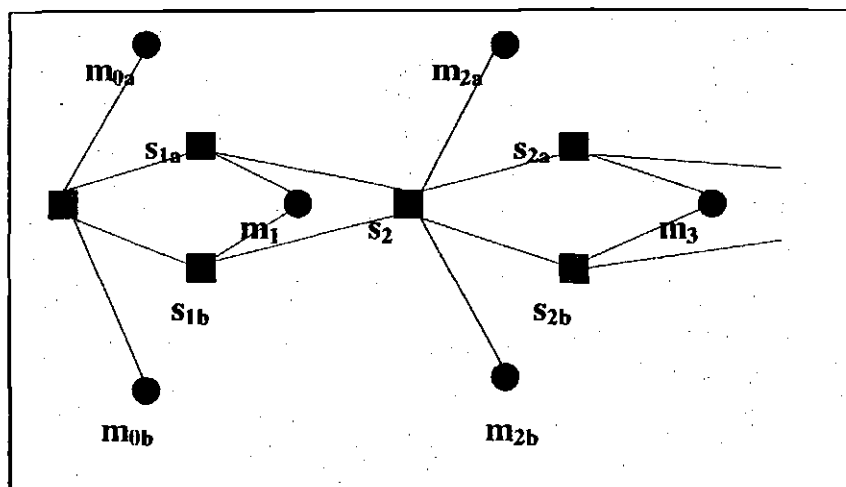


Figure 3. Schematic view of blast pattern often used in Platinum mining.

Thus, for any given values of μ_s , σ_s and σ_m the corresponding value of θ can be accurately estimated via Monte Carlo simulation, as

indicated in section 1 of the appendix. Tables 6a and 6b show the pairwise and sequence length 100 reversal probabilities calculated for the same parameter configuration used in tables 1a and 2a. It is notable how much more (less) sensitive reversal probabilities in the diamond-shaped configuration are to changes in the surface (main) delay standard deviation.

Table 6a. Pairwise reversal probabilities θ for $\mu_s = 200$ and various combinations of σ_s and σ_m in the diamond-shaped configuration.

		σ_m			
		30	40	50	60
σ_s	4	4.7×10^{-6}	5.6×10^{-6}	7.0×10^{-6}	9.1×10^{-6}
	6	9.2×10^{-4}	9.9×10^{-4}	1.1×10^{-3}	1.2×10^{-3}
	8	9.9×10^{-3}	1.0×10^{-2}	1.1×10^{-2}	1.1×10^{-2}
	10	3.7×10^{-2}	3.7×10^{-2}	3.9×10^{-2}	4.0×10^{-2}

Table 6b. Reversal probabilities Θ for $\mu_s = 200$ and sequence length $n = 100$ in the diamond-shaped configuration.

		σ_m			
		30	40	50	60
σ_s	4	4.7×10^{-4}	5.6×10^{-4}	7.0×10^{-4}	9.1×10^{-4}
	6	8.8×10^{-2}	9.4×10^{-2}	1.0×10^{-1}	1.1×10^{-1}
	8	6.3×10^{-1}	6.4×10^{-1}	6.6×10^{-1}	6.8×10^{-1}
	10	9.8×10^{-1}	9.8×10^{-1}	9.8×10^{-1}	9.8×10^{-1}

8. Acknowledgements

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10. Appendix

1. (Section 2.1) The Bonferroni inequalities (Feller, 1968, page 110) are

$$S_1 - S_2 + \cdots - S_{2r} \leq \Pr\{R\} \leq S_1 - S_2 + \cdots + S_{2r-1} \quad (13)$$

for $r = 1, 2, \dots, n$ where

$$S_r = \sum_{k_1 < \dots < k_r} \Pr\{A_{k_1} \cap \dots \cap A_{k_r}\}.$$

Writing

$$\theta_l = \Pr\{A_1 \cap \dots \cap A_l\}, \quad l = 1, 2, \dots$$

and using the fact that A_l and A_m are independent whenever $|l - m| \geq 2$ we find

$$S_1 = n\theta_1$$

$$S_2 = \binom{n-1}{2}\theta_1^2 + (n-1)\theta_2,$$

$$S_3 = \binom{n-2}{3}\theta_1^3 + 2\binom{n-2}{2}\theta_1\theta_2 + (n-2)\theta_3,$$

$$S_4 = \binom{n-3}{4}\theta_1^4 + 3\binom{n-3}{3}\theta_1^2\theta_2 + \binom{n-3}{2}(2\theta_1\theta_3 + \theta_2^2) + (n-3)\theta_4,$$

etcetera. Note that θ_1 equals θ defined in (2.1). Conditioning on $m_1, m_2, \dots$ gives

$$\theta_2 = E\{\Phi((- \sigma_m Z_1 - \mu_s)/\tau)\Phi((\sigma_m Z_1 - \mu_s)/\tau)\} \quad (14)$$

and, for $l \geq 3$,

$$\theta_l = E\left\{\Phi((- \sigma_m Z_1 - \mu_s)/\tau)\Phi((\sigma_m Z_{l-1} - \mu_s)/\tau) \times \prod_{j=1}^{l-2} \Phi((\sigma_m Z_j - \sigma_m Z_{j+1} - \mu_s)/\sigma_s)\right\}$$

where $Z_1, Z_2, \dots$ are independent $N(0, 1)$ random variables and $\tau = (\sigma_m^2 + \sigma_s^2)^{1/2}$ (see appendix 4 below for a detailed calculation in a similar, slightly more complicated, setup). For any given values of μ_s , σ_m and σ_s the θ_l can now be evaluated by numerical integration or can be accurately estimated by Monte Carlo simulation. For instance, if the random variable

over which the expectation is taken in (14) is denoted by $g(Z_1)$; and if $N_1, \dots, N_B$ denote B standard normal random numbers, $\hat{\theta}_2 = B^{-1} \sum_{i=1}^B g(N_i)$ is an unbiased estimate of θ_2 with standard deviation $\leq (\theta_2/B)^{1/2}$. Hence, $S_2, S_3, \dots$ can be evaluated until $\Pr\{R\}$ is determined to any desired accuracy. For each of the parameter configurations displayed in tables 2a and 2b the value of $1 - (1 - \theta)^n$ lay between the lower and upper bounds of an inequality in (13) in which the error, rounded to the last significant digit given in the tables, was zero.

It is perhaps worthwhile noting that the approximation

$$\Pr\{R\} \approx 1 - (1 - \theta)^n$$

has a theoretical basis in the extreme value behaviour of the stationary stochastic process (see (3))

$$X_j = T_j - T_{j-1} = s_j + m_j - m_{j-1}, \quad j \geq 1,$$

of time lapses between explosion of successive in-hole detonators. Notice that the event R occurs if and only if the event $\{\min_{1 \leq j \leq n} X_j < 0\}$ occurs. Also, the process $\{X_j\}$ has a short memory because X_j and $X_{j'}$ are independent whenever $|j - j'| \geq 2$; see (12). Hence, for "large" n , the minimum $\min_{1 \leq j \leq n} X_j$ behaves more or less as it would have had the X_j been independent; see section 9.5 of David (1981). Thus

$$\begin{aligned} \Pr\{R\} &= 1 - \Pr\{\min_{1 \leq j \leq n} X_j > 0\} \\ &\approx 1 - (\Pr\{X_1 > 0\})^n \\ &= 1 - (1 - \theta)^n. \end{aligned}$$

2. (Section 2.2) For $1/2 < \gamma < 1$ the $100\gamma\%$ and $100(1-\gamma)\%$ quantiles of B_0 are approximately given by the solutions of the equations

$$\Phi\left((\mu_m - Y\mu_s)/(\sigma_m^2 + Y\sigma_s^2)^{1/2}\right) = \gamma, \quad 1 - \gamma,$$

namely

$$Y(\gamma) \approx \mu_m/\mu_s + |z_\gamma| [\sigma_s^2\mu_m/\mu_s^3 + \sigma_m^2/\mu_s^2]^{1/2}$$

and

$$Y(1-\gamma) \approx \mu_m/\mu_s - |z_\gamma| [\sigma_s^2\mu_m/\mu_s^3 + \sigma_m^2/\mu_s^2]^{1/2}$$

respectively, where we use the fact that $z_\gamma = -z_{1-\gamma}$ and assume that $\sigma_s^2/\mu_s^2 \approx 0$. (The latter approximation is justified because σ_s^2/μ_s^2 is typically at least one order of magnitude smaller than μ_m/μ_s , $\sigma_s^2\mu_m/\mu_s^3$ and σ_m^2/μ_s^2 .) These approximations suggest that the distribution of B_0 is near to being symmetric around μ_m/μ_s . Also,

$$Y(\gamma) - Y(1-\gamma) = 2|z_\gamma| [\sigma_s^2\mu_m/\mu_s^3 + \sigma_m^2/\mu_s^2]^{1/2}$$

which, upon taking $|z_\gamma| = 3$, indicates that the distribution of B_0 is concentrated effectively in a range of width $6w$ around the median, where

$$w = (\sigma_s^2\mu_m/\mu_s^3 + \sigma_m^2/\mu_s^2)^{1/2}. \quad (15)$$

Since w never exceeds $1/2$ in realistic cases, it is to be expected that the distribution of B_0 will effectively be concentrated on two or three integers which are more or less symmetrically situated around μ_m/μ_s .

The same qualitative conclusions hold when the assumption that all distributions are normal is not satisfied. This claim can be justified on the basis of the approximations

$$E\{\mathbf{I}\} \approx \mu_m/\mu_s - 1/2$$

and

$$\text{var}\{I\} < (0.58)^2$$

which follow upon applying results from renewal theory to the sequence $s_1, s_2, \dots$ of positive, independent and identically distributed, random variables. Fix m_0 . Then I is the number of renewals in the interval $(0, m_0]$. If m_0 is "large", we have the following approximations for the conditional expectation and variance of I , namely

$$E\{I|m_0\} \approx m_0/\mu_s - 1/2 \text{ and } \text{var}\{I|m_0\} \approx m_0\sigma_s^2/\mu_s^3 + 1/12;$$

see Cox (1970, pages 47 and 58). Then

$$E\{I\} = E\{E\{I|m_0\}\} \approx E\{m_0/\mu_s - 1/2\} = \mu_m/\mu_s - 1/2$$

and

$$\begin{aligned} \text{var}\{I\} &= E\{\text{var}\{I|m_0\}\} + \text{var}\{E\{I|m_0\}\} \\ &\approx E\{m_0\sigma_s^2/\mu_s^3\} + \text{var}\{m_0/\mu_s\} + 1/12 \\ &= \mu_m\sigma_s^2/\mu_s^3 + \sigma_m^2/\mu_m^2 + 1/12 \\ &= w^2 + 1/12 \\ &< 1/4 + 1/12 = (.58)^2 \end{aligned}$$

because w , given in (15), is typically less than $1/2$.

3. (Section 4) The quantity x in (4) is a function of three variables. Application of the delta method (Rao, 1965, pp. 319–322) yields that $\hat{x}$ in (9) is, for large k and l , approximately normally distributed with mean x and variance

$$(\sigma_s/\mu_s)^2 x^2/k + (\sigma_s/\mu_s)^4 x^6/2k + 2(\sigma_m/\mu_s)^4 x^6/l. \quad (16)$$

In typical parameter configurations $\hat{\sigma}_s/\hat{\mu}_s$ is an order of magnitude less than $\hat{\sigma}_m/\hat{\mu}_s$ so that the first two terms in (16) are negligible relative to the third. Furthermore, $x = -(\mu_s/\sigma_m) / (\sigma_s^2/\sigma_m^2 + 2)^{1/2} \approx -(\mu_s/\sigma_m)/2^{1/2}$ because σ_s^2/σ_m^2 is typically negligible compared to 2. Hence, $\sigma_m/\mu_s \approx 2^{1/2} \times x^{-1}$. Substitution of these approximations into (16) yields $x^2 \times (2l)^{-1}$ for the approximate variance of $\hat{x}$.

4. (Section 7) Since the C_k 's all have the same probability it suffices to find an expression for $\Pr\{C_1\}$. We have $T_{0a} = m_{0a}$, $T_{0b} = m_{0b}$, $T_1 = \min(s_{1a}, s_{1b}) + m_1$ and $T_{2a} = \min(s_{1a}, s_{1b}) + s_2 + m_{2a}$, $T_{2b} = \min(s_{1a}, s_{1b}) + s_2 + m_{2b}$. Let $Z_0, Z_{1a}, Z_{1b}, Z_1, Z_2$ and Z_3 be independent $N(0, 1)$ random variables and note that we can write $s_{1i} = \mu_s + \sigma_s Z_{1i}$ for $i = a$ or b , $s_2 = \mu_s + \sigma_s Z_2$, $m_0 = \mu_m + \sigma_m Z_0$, $m_1 = \mu_m + \sigma_m Z$ and $m_{2a} = \mu_m + \sigma_m Z_3$. Conditioning on m_1, s_{1a}, s_{1b} and s_2 one sees that

$\Pr\{C_1\}$ equals

$$\begin{aligned} & E[\Pr\{\max(m_{0a}, m_{0b}) < \min(s_{1a}, s_{1b}) + m_1 | m_1, s_{1a}, s_{1b}\} \\ & \quad \times \Pr\{m_1 < s_2 + \min(m_{2a}, m_{2b}) | m_1, s_2\}] \\ & = E[\Pr\{m_{0a} < \min(s_{1a}, s_{1b}) + m_1, m_{0b} < \min(s_{1a}, s_{1b}) + m_1 | m_1, s_{1a}, s_{1b}\} \\ & \quad \times \Pr\{m_{2a} > m_1 - s_2, m_{2b} > m_1 - s_2 | m_1, s_2\}] \\ & = E[\Pr^2\{\sigma_m Z_0 < \mu_s + \sigma_s \min(Z_{1a}, Z_{1b}) + \sigma_m Z_1 | Z_1, Z_{1a}, Z_{1b}\} \\ & \quad \times \Pr^2\{\sigma_m Z_3 > \sigma_m Z_1 - \mu_s - \sigma_s Z_2 | Z_1, Z_2\}] \\ & = E[\Phi^2(\mu_s/\sigma_m + \min(Z_{1a}, Z_{1b})\sigma_s/\sigma_m + Z_1) \\ & \quad \times \Phi^2(\mu_s/\sigma_m + Z_2\sigma_s/\sigma_m - Z_1)]. \end{aligned}$$

